



period of a string

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Given a string S , find the smallest positive length P such that the string can be formed by repeating its prefix of length P

Input Format

A single line containing the string S .

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Output Format

Print the smallest period of the string.

Sample Input 0

```
ababab
```

Sample Output 0

```
2
```

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Language

Python 3



```
1 import sys
2
3 def find_smallest_period():
4     input_data = sys.stdin.read().split()
5     if not input_data:
6         return
7
8     s = input_data[0]
9     n = len(s)
10
11
12     lps = [0] * n
13     length = 0
14     i = 1
15
16     while i < n:
17         if s[i] == s[length]:
18             length += 1
19             lps[i] = length
20             i += 1
21         else:
22             if length != 0:
23                 length = lps[length - 1]
24             else:
```

```
25         lps[i] = 0
26         i += 1
27
28     border_len = lps[-1]
29     candidate_period = n - border_len
30
```

Line: 38 Col: 1

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Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

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```
1 | ababab
```

Your Output (stdout)

```
1 | 2
```

Expected Output

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```
1 | 2
```